CRASH COURSE

— GATE 2026 —

COMPUTER SCIENCE & IT

Computer Networks 05

Transport Layer
[TCP Congestion Control]

By - Abhishek Sir

Recap of previous lecture

1. TCP Connection Establishment → V. Imp.
2. TCP Data Transfer

Topics to be covered

1. TCP Flow & Congestion Control Mechanism
2. TCP Congestion Control Algorithm → AIMD

V.V. Imp.

(CS-24), CS-20, CS-12, CS/IT-04, CS-14

ABOUT ME

Hello, I'm Abhishek

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Topic : Reference

→ Request for Comments (RFC)

by Internet Engineering Task Force (IETF)

Topic : Syllabus

Transport layer : flow control and congestion control, UDP, TCP, sockets;

Topic : TCP Header

<table>
<tr><td>0</td><td></td><td></td><td>16</td><td>31</td></tr>
<tr><td colspan="3">Source Port Number (16 bit)</td><td colspan="2">Destination Port Number (16 bit)</td></tr>
<tr><td colspan="5">Sequence Number (32 - bits)</td></tr>
<tr><td colspan="5">Acknowledgment Number (32 - bits)</td></tr>
<tr><td>HLEN</td><td>Reserved
Must be zero</td><td>6 Flag bits</td><td colspan="2">Window Size (16 bit)</td></tr>
<tr><td colspan="3">Checksum (16 bit)</td><td colspan="2">Urgent Pointer (16 bit)</td></tr>
<tr><td colspan="5">Optional Header</td></tr>
<tr><td colspan="5">Payload</td></tr>
</table>

Topic : TCP


TCP Client Process
P_1
SOCK_STREAM
(TCP Socket)
Send Buffer
Recv Buffer
TCP Server Process
P_2
SOCK_STREAM
(TCP Socket)
Send Buffer
Recv Buffer

Topic : TCP Buffer

→ Each TCP socket have two buffers :-

1. Send buffer size : MaxSendBuffer
2. Receive buffer size : MaxRcvBuffer

TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

LastByteWritten = P_1's ISN

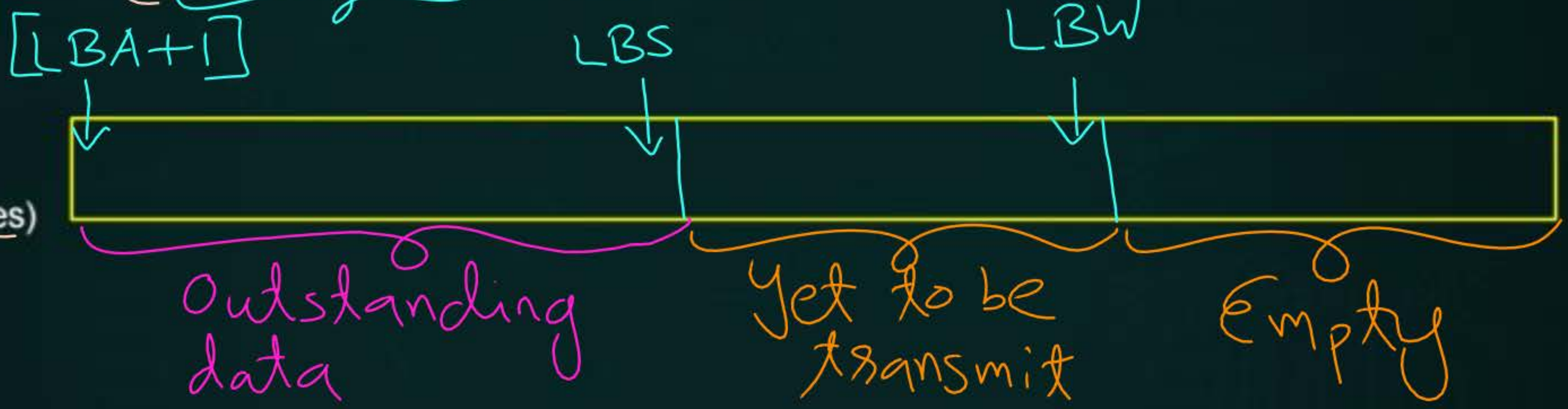
LastByteSent = P_1's ISN

LastByteAcked = P_1's ISN

RWND =

CWND =

Send Buffer
(MaxSendBuffer = 1000 Bytes)



Outstanding (inflight) ⇒ Transmitted data, yet to be acked.

TCP Server Process

P_2

SOCK_STREAM
(TCP Socket)

Recv Buffer
(MaxRecvBuffer = 1200 Bytes)

LastByteRead = P_1's ISN

NextByteExpected = P_1's ISN + 1

LastByteRcvd = P_1's ISN

Topic : Window Size

→ Receiver Window Size (16-bit)

→ An advertisement from receiver

[Receiver Advertised Window Size]

Topic : Window Size

→ RWND : Receiver Advertised Window Size

→ Define how much data (in bytes) the receiving device can accept

[Define current available space (in bytes) in receiver's receiving window]

Topic : TCP Flow & Congestion Control

$$\underbrace{[\text{LastByteSent} - \text{LastByteAcked}]}_{\text{current outstanding data}} \leq \underbrace{\text{minimum}[\text{CWND}, \text{RWND}]}_{\text{Max}^m \text{ outstanding data can be.}}$$

[TCP Cong. control Algo.]

CWND : Sender's Congestion Window Size [Only for congestion control]

RWND : Receiver's advertised available space in Receiving Window [Only for flow control]

Topic : TCP Flow & Congestion Control

Mech.

=> TCP sender wants to send a segment contains N bytes in its payload

if [(LastByteSent – LastByteAcked) + N] ≤ minimum[CWND, RWND]

→ Allowed to send.

else

→ Not allowed to send immediately, wait for sometime.

Topic : TCP Congestion Control Algorithm

→ AIMD : Additive Increase Multiplicative Decrease

[TCP Congestion Control Algorithm]

AIAD
MIMD
MIMD

$\boxed{\text{AIMD}}$ ←

Topic : TCP Congestion Control Algorithm

→ Two phase of TCP Congestion Control Algorithm :-

1. Slow Start Phase

2. Congestion Avoidance Phase

Topic : TCP Congestion Control Algorithm

→ Two phase of TCP Congestion Control Algorithm :-

1. Slow Start Phase

   [when CWND < SSThresh]

2. Congestion Avoidance Phase

   [when CWND ≥ SSThresh]

RFC

SSThresh : Slow Start Threshold

Topic : Maximum Segment Size

→ Maximum Segment Size (MSS)

→ Maximum TCP segment payload size (in bytes)

[Maximum bytes can be kept in the payload field of a TCP segment]

Max^m TCP seg. size = [TCP Header Size + MSS]

CWND → in MSS [Algo. wise]
└→ (int)

CWND → in bytes (Prog. wise)
└→ (int)

Topic : Slow Start Phase

→ When Congestion Window Size is less then Slow Start Threshold

CWND < SSThresh

→ CWND increase by 1 MSS on every successful acknowledgment

$$\boxed{CWND = [CWND + 1];}$$

Topic : Slow Start Phase

Suppose $\underbrace{RWND = 100\ MSS}$

$\underbrace{\text{SSThresh}} = \underbrace{\text{Infinite}}$ ✓

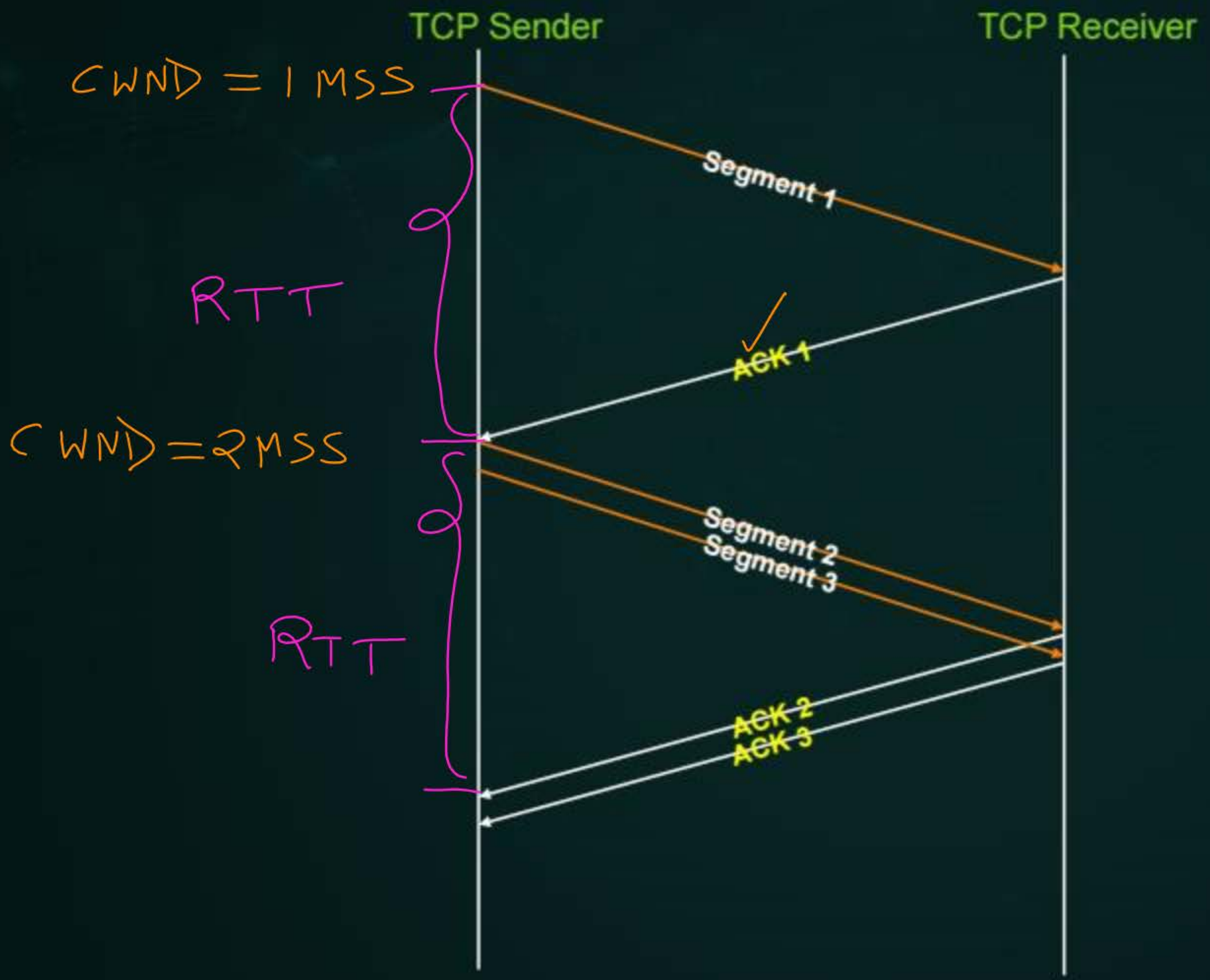
$\underbrace{\text{CWND}} = \underbrace{1\ MSS}$

2MSS

4MSS

8MSS

16MSS


TCP Sender
TCP Receiver
CWND = 1 MSS
Segment 1
RTT
ACK 1
CWND = 2 MSS
Segment 2
Segment 3
RTT
ACK 2
ACK 3

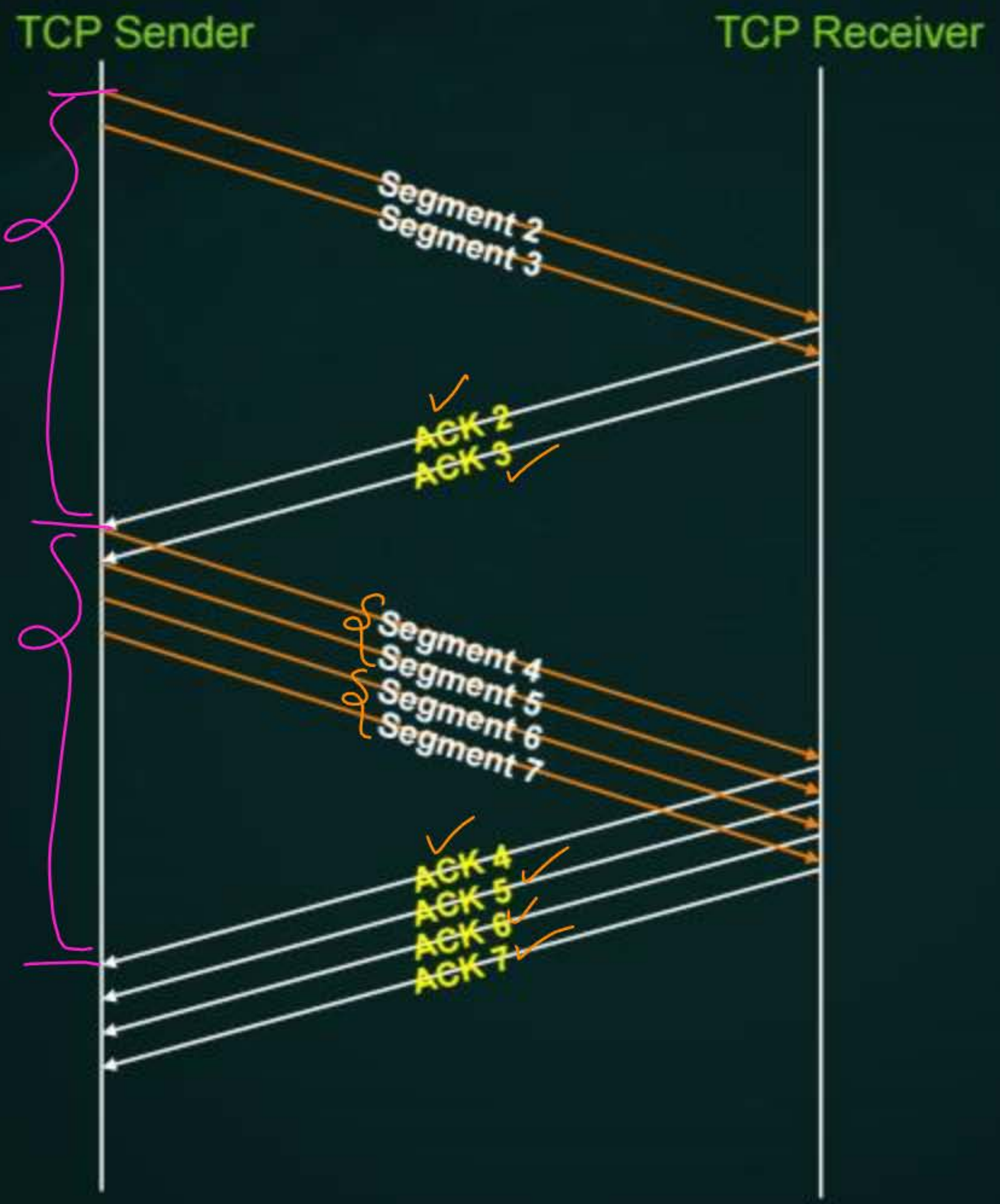
CWND=2MSS

2+1=3

(3+1=4)

RTT

CWND=4MSSS



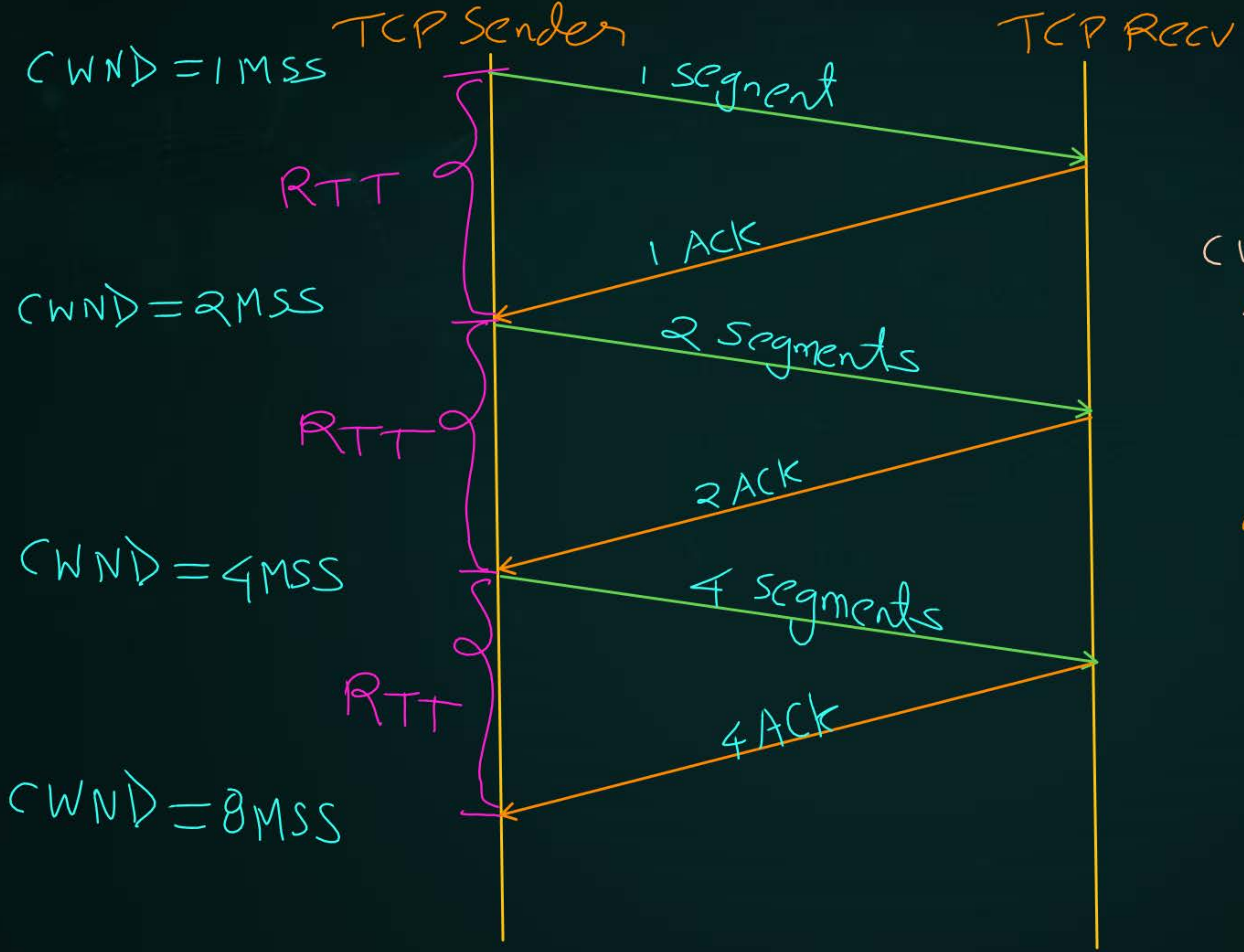
(4+1)

RTT

(5+1)

(6+1)

(7+1) CWND=8MSS



TCP Sender
TCP Recv
CWND = 1 MSS
1 segment
RTT
1 ACK
CWND = 2 MSS
2 Segments
RTT
2 ACK
CWND = 4 MSS
4 segments
RTT
4 ACK
CWND = 8 MSS

CWND
0 1 2 3 4 5
RTT

Topic : Slow Start Phase

→ CWND approximately doubles on every round trip time (RTT)

→ Congestion Window (CWND) grow exponentially

[MCQ] [GATE-CS-2008, 2-Marks] IISC

#Q. In the slow start phase of the TCP congestion control algorithm, the size of the congestion window :

(A) ~~does not increase~~

(B) ~~increases linearly~~

(C) ~~increases quadratically~~

(D) increases exponentially ✓

$\boxed{\text{Ans: D}}$

[MCQ] [GATE-2018, 1-Mark] IIT-G,

#Q. Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that cwnd stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

(1 MSS)

i. The cwnd increase by 2 MSS on every successful acknowledgement. False

ii. The cwnd approximately doubles on every successful acknowledgedment. False

iii. The cwnd increase by 1 MSS every round trip time. False

iv. The cwnd approximately doubles every round trip time. True

Which one of the following is correct ?

A. Only (ii) and (iii) are true

B. Only (i) and (iii) are true

C. Only (iv) is true

D. Only (i) and (iv) are true

$\boxed{\text{Ans: C}}$

Topic : Congestion Avoidance Phase

→ When Congestion Window Size is greater then equal to Slow Start Threshold

$CWND \geq SSThresh$

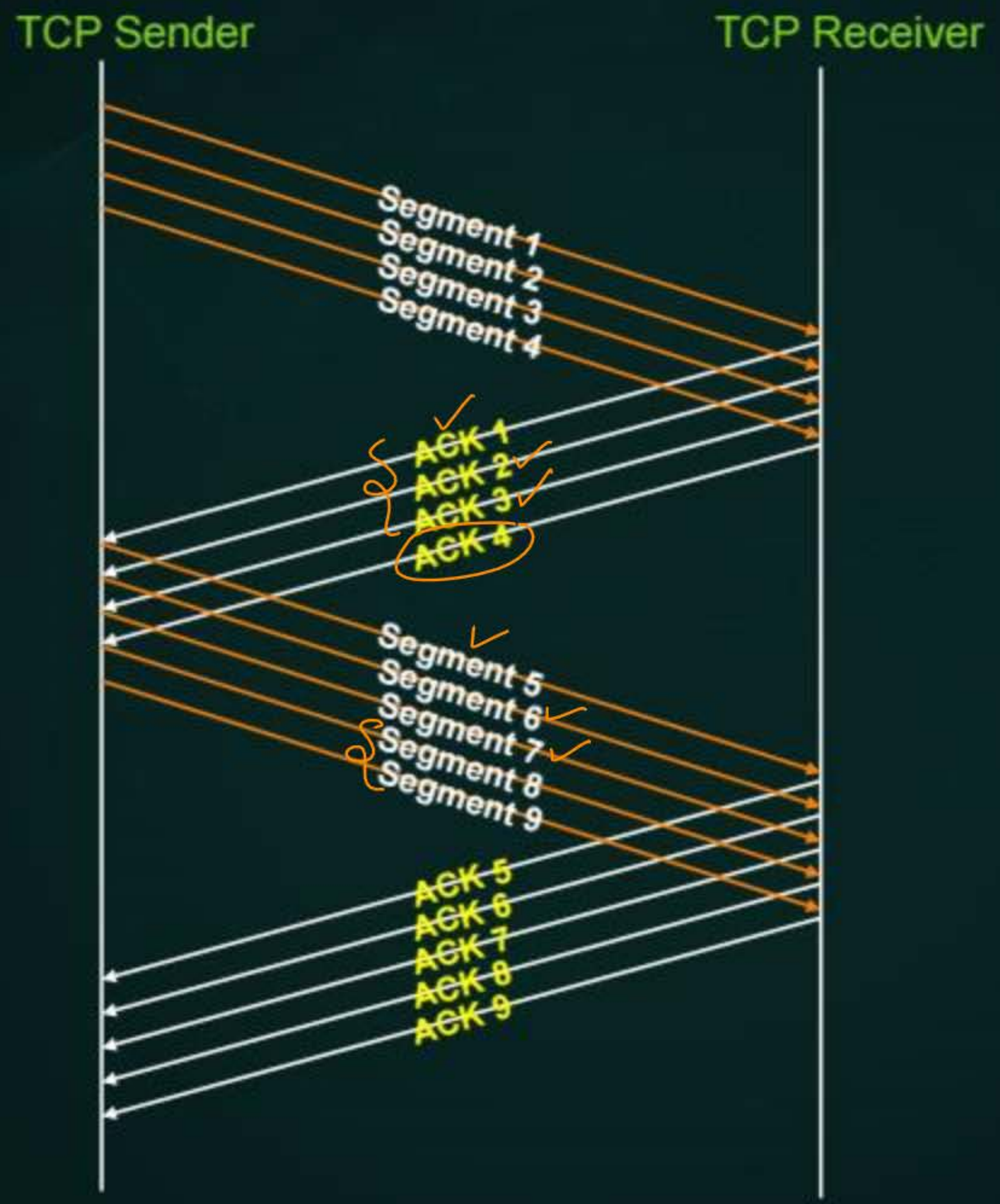
→ CWND increase by 1 MSS on every round trip time ✓ → Additive Increase

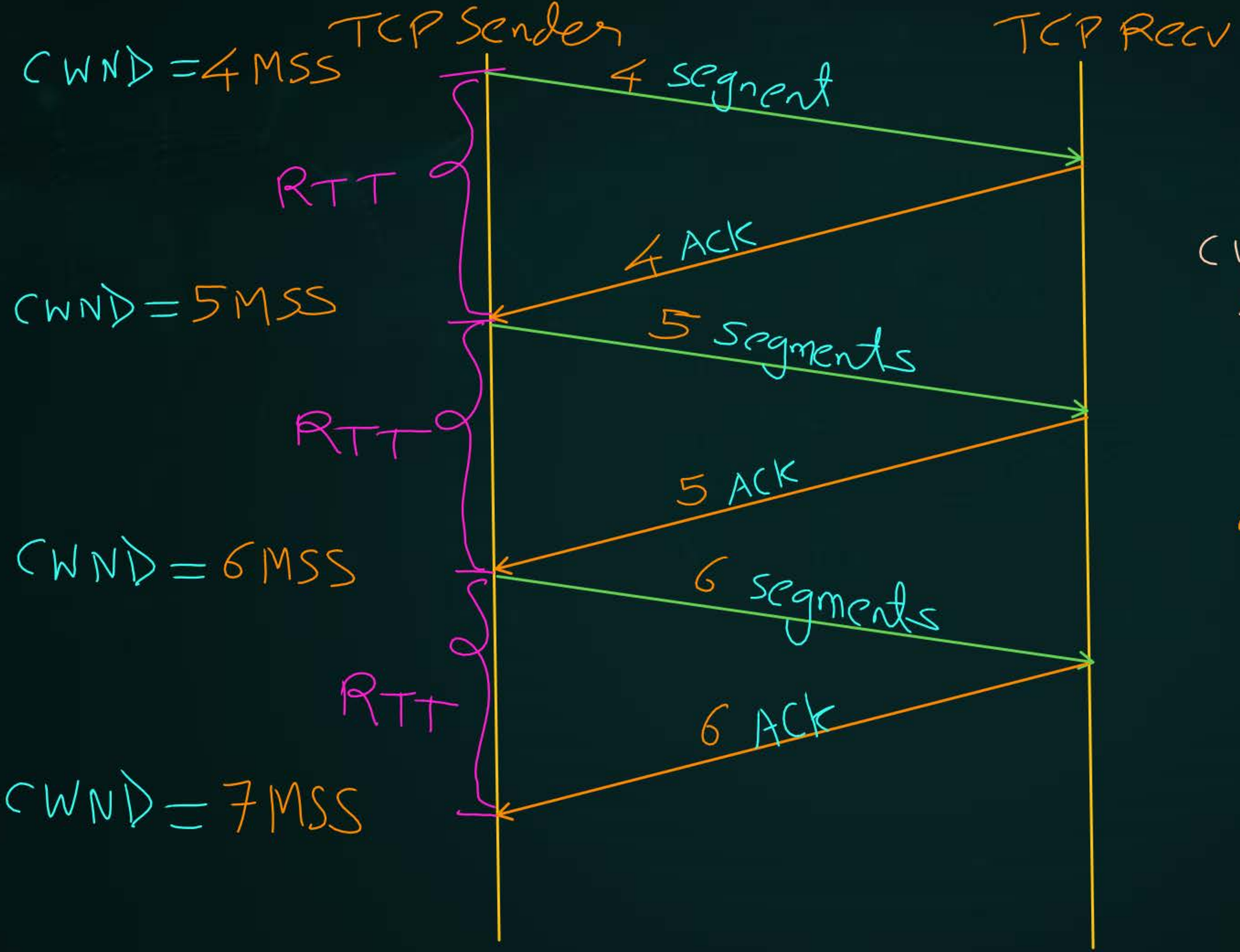
[On successful acknowledgments of all transmitted segments]

Topic : Congestion Avoidance Phase

SSThresh = 4 MSS

CWND = 4 MSS


TCP Sender
TCP Receiver
Segment 1
Segment 2
Segment 3
Segment 4
ACK 1
ACK 2
ACK 3
ACK 4
Segment 5
Segment 6
Segment 7
Segment 8
Segment 9
ACK 5
ACK 6
ACK 7
ACK 8
ACK 9



TCP Sender
TCP Recv
CWND = 4 MSS
4 segment
RTT
4 ACK
CWND = 5 MSS
5 segments
RTT
5 ACK
CWND = 6 MSS
6 segments
RTT
6 ACK
CWND = 7 MSS

CWND
0 1 2 3 4 5
RTT

Topic : Congestion Avoidance Phase

→ Congestion Window (CWND) grow linearly

Topic : TCP Congestion Control Algorithm

→ When TCP connection established :

large no.

Initialize :	SSThresh	=	Infinite
	CWND	=	1 MSS

$= \boxed{(2^{16}-1) * 2^{14}}$ ⟹ Max^{m} window size can be advertized by TCP Recv. using 'window scale'.

→ by default

As per programmer
$\underbrace{[2\text{ MSS}, 3\text{ MSS}, 4\text{ MSS} \ldots]}_{\text{lower no.}}$

Topic : TCP Congestion Control Algorithm

→ When Time-out occur :

Update : SSThresh = [CWND / 2]

CWND = 1 MSS

⟹ $\left[\dfrac{\text{No. of outstanding data segments (inflight data)}}{2}\right]$

⟹ As per prog. ceil or Floor

Multiplicative Decrease

SSThresh = Infinite

CWND = 1 MSS

2MSS

4MSS

8MSS

CWND = 16MSS

Time-out

slow start phase

$SSThresh = \frac{CWND}{2} = \frac{16MSS}{2} = 8MSS$

CWND = 1 MSS

2MSS

4MSS

8MSS

slow start phase

9MSS

10MSS

11MSS

⋮

Cong. Avoidence phase

Example 12 :- [NAT]

#Q. Consider a TCP connection at some point of time when the congestion window size is 16 MSS (Maximum Segment Size), timeout occurs due to packet loss. How many RTTs (Round Trip Time) the TCP connection takes to enters into congestion avoidance phase after timeout?

$\boxed{\text{Ans} = 3 \text{ to } 4}$

CWND = 16 MSS

*

Time-out

SSthresh = CWND/2

$= \frac{16\text{ MSS}}{2}$

SSthresh = 8 MSS

SSthresh = 8 MSS

CWND = 1 MSS

1

2MSS

2

4MSS

3

8MSS

*

9MSS

slow start phase

[NAT] [GATE-2024, Set-2, 2-Marks] IISc

#Q. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _______ .

$\boxed{\text{Ans} = 4}$

CWND = 12 MSS

Time-out

SSthresh = CWND/2

$= \frac{12 \text{ MSS}}{2}$

SSthresh = 6 MSS

CWND = 1 MSS

① RTT

2 MSS

② RTT

CWND = 4 MSS

③ RTT

[B/W 6 to 7 MSS]

Example 13 :-

[NAT]

#Q. Consider a TCP connection at some point of time when the congestion window size is 16 MSS (Maximum Segment Size), timeout occurs due to packet loss. How many RTTs (Round Trip Time) the TCP connection takes to get back to 16 MSS congestion window size after timeout?

$\boxed{\text{Ans} = 11 \text{ to } 12}$

CWND = 16 MSS *

Time-out

SSthresh = CWND/2

$= \frac{16\,MSS}{2}$

SSthresh = 8 MSS

SSthresh = 8 MSS

CWND = 1 MSS

1

2MSS

2

4MSS

3

8MSS

4

9MSS

5

10MSS

⋮

15MSS

11

16MSS

slow start phase

[NAT] [GATE-2014, Set-1, 2-Marks] IIT-KGP, H.W.

#Q. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is ______ .

[NAT] [GATE-2020, 2-Marks] IIT-D, H.W.

#Q. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time t=0. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time t+60 ms after all acknowledgments are processed is _______ .

[MCQ] [GATE-2012, 2-Marks] IIT-D, H.W.

#Q. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

(A) 7 MSS

(B) 8 MSS

(C) 12 MSS

(D) 14 MSS

IIT-D $\boxed{2004}$

$\boxed{\text{Ans: IIT-D Marks to All}}$

Summary

1. TCP Flow & Congestion Control Mechanism
2. TCP Congestion Control Algorithm → AIMD

THANK - YOU